farmed articles[5], putting more emphasis on content produced by trustworthy sources *e.g.* education domains, while DuckDuckGo removed them altogether[6].

What is different with the arrival of LLMs is the scale at which such poisoning can happen once it is automated. Preserving the ability of LLMs to model low-probability events is essential to the fairness of their predictions: such events are often relevant to marginalised groups. Low-probability events are also vital to understand complex systems [Taleb, 2007].

Our evaluation suggests a "first mover advantage" when it comes to training models such as LLMs. In our work we demonstrate that training on samples from another generative model can induce a distribution shift, which over time causes *Model Collapse*. This in turn causes the model to mis-perceive the underlying learning task. To make sure that learning is sustained over a long time period, one needs to make sure that access to the original data source is preserved and that additional data not generated by LLMs remain available over time. The need to distinguish data generated by LLMs from other data raises questions around the provenance of content that is crawled from the Internet: it is unclear how content generated by LLMs can be tracked at scale. One option is community-wide coordination to ensure that different parties involved in LLM creation and deployment share the information needed to resolve questions of provenance. Otherwise, it may become increasingly difficult to train newer versions of LLMs without access to data that was crawled from the Internet prior to the mass adoption of the technology, or direct access to data generated by humans at scale.

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A Appendix

A.1 Absorbing Markov Chain

The subsection explains a well-known fact about absorbing Markov chains, that they converge to an absorbing state with probability one. Assume that $\mathbf{X}^m$ form a Markov chain. In order to reason about this chain we need to consider the transition probabilities. In general, these correspond to our functional approximation scheme. Due to the stochastic nature of the Markov chain, we expect to have the variance go up and down. But as the variance decreases, the newly sampled data, due to its finiteness, will be more concentrated, leading in the limit to a set of *i.e.* a delta functions. This argument assumes that the approximation scheme is good and can converge to delta functions. If not, the errors in approximation may prevent the propagation of errors in stochasticity.

As discussed in the previous section, we can model the process of repeated 'sampling' and 'fitting' as a Markov chain. In this subsection, we explain how such a process can converge to a stationary state *i.e.* the absorbing state of a Markov Chain. In this derivation we follow Allan Yashinski [7]. Suppose we have an absorbing Markov Chain with r transient states $t_1, \dots, t_r$ and s absorbing states $a_1, \dots, a_s$. The whole Markov chain has $r + s$ states, ordered as follows: $t_1, \dots, t_r, a_1, \dots, a_s$. The transition matrix is then defined as

$$T = \begin{bmatrix} Q & 0_{r\times s} \\ R & I_s \end{bmatrix}, \tag{16}$$

where

- Q is an $r \times r$ matrix holds the probabilities of moving from a transient state to another transient state
- R is an $s \times r$ matrix which holds the probabilities of moving from a transient state to an absorbing state.
- $0_{r\times s}$ is the $r \times s$ matrix of all 0's. There 0's represent the probabilities of moving from an absorbing state to a transient state (which is impossible by definition).
- I_s holds the probabilities of transitioning between the absorbing states. As transition is impossible, this is just the $s \times s$ identity matrix.

We are interested in $\lim_{k\to\infty} T^k(\mathbf{X}_0)$. For a given k, the matrix becomes

$$T^k = \begin{bmatrix} Q^k & 0_{r\times s} \\ R + RQ + \cdots + RQ^{k-1} & I_s \end{bmatrix} = \begin{bmatrix} Q^k & 0_{r\times s} \\ R\sum_{i=0}^{k-1} Q^i & I_s \end{bmatrix}. \tag{17}$$

Finally, for an absorbing Markov chain with $T = \begin{bmatrix} Q & 0_{r\times s} \\ R & I_s \end{bmatrix}$,

we have $\lim_{k\to\infty} T^k = \begin{bmatrix} 0_{r\times r} & 0_{r\times s} \\ R(I_r - Q)^{-1} & I_s \end{bmatrix}$.

Since in the limit the transition probabilities to transient states are zero, we end up converging to absorbing states and staying there. In the case of discrete distributions, where we can perfectly approximate a zero-variance dataset (*i.e.* a delta function), the absorbing states are delta functions centered at any non-zero probability point from the original distribution. In practice, we would like to know the expected number of steps before being absorbed, which may be large. But without knowing our fitting procedure it is impossible to calculate the matrix Q and therefore the average length of time before collapse.

A.2 Alternative assumption for noisy approximations

This subsection will cover an alternative assumption, which may be more realistic in **some** settings, in contrast to assumption 3 from Section 4.3, and this subsection mostly acts as an extension, rather than an alternative. In particular, instead of imposing orthogonality, we can instead impose a certain size requirement on the noise term. This in turn allows us to arrive to a similar result.

To be more precise, we will consider the same setting as in Section 4.3, but we will now replace Assumption 3 with Assumption 3*:

[7] www.math.umd.edu/~immortal/MATH401/book/ch_absorbing_markov_chains.pdf